DAY 8 | EXCEL

My Data Analyst Journey

Math Functions:
FLOOR, CEILING, MOD & More

Day 8 of learning Excel in public.

ABS **MOD** **^**

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The Idea: 50 / 3

Why FLOOR, CEILING and MROUND exist

The Problem

50 / 3 = 16.666... It does not divide evenly.

FLOOR

Goes down to the next multiple of 3: 48. Then 48 / 3 = 16.

CEILING

Goes up to the next multiple of 3: 51. Then 51 / 3 = 17.

MROUND

Picks the nearest multiple: 51 (1 away) beats 48 (2 away).

FLOOR, CEILING, MROUND

=FUNCTION(number, multiple)

```
=FLOOR(50,3)
```
48

```
=CEILING(50,3)
```
51

```
=MROUND(50,3)
```
51

```
=MROUND(49,3)
```
48

The second number is the multiple you want. The answer is always a multiple of it. FLOOR.MATH and CEILING.MATH also handle negative numbers.

ABS

Remove the negative sign

`=ABS(-10)`	10
`=ABS(10)`	10
`=-ABS(10)`	-10
`=ABS(A1-B1)`	Gap between two values

ABS turns a negative into a positive. A minus sign in front of ABS forces the answer to be negative. Great for differences where order does not matter.

MOD

Gives the remainder

```
=MOD(10,3)
```
1

```
=MOD(9,3)
```
0

```
=MOD(7,2)
```
1

```
=QUOTIENT(10,3)
```
3

10 / 3 = 3 remainder 1. MOD gives the remainder. QUOTIENT gives only the whole-number part. MOD = 0 means the number divides evenly.

MOD in Real Life

Four handy uses

Even or Odd

=MOD(A1,2)=0 gives TRUE for even numbers.

Every 3rd Row

=MOD(ROW(),3)=0 marks rows 3, 6, 9... Great with Conditional Formatting.

Divisible?

=MOD(A1,5)=0 checks if a number is a multiple of 5.

Time Split

=QUOTIENT(135,60) = 2 hours and =MOD(135,60) = 15 minutes.

POWER & SQRT

Powers and square roots

`=POWER(2,3)`	8
`=2^3`	8
`=POWER(5,2)`	25
`=SQRT(16)`	4
`=SQRT(50)`	7.0710678

2 to the power 3 means 2 x 2 x 2 = 8 (not 6). SQRT of a negative number gives #NUM!. Use SQRT(ABS(x)) if the sign is unknown.

Combine Two Formulas

Nesting: one function inside another

```
=ROUND(SQRT(50),2)
```
7.07

```
=ROUND(SQRT(K33),2)
```
Root first, then round

```
=SQRT(ABS(-16))
```
4

```
=MOD(ABS(-10),3)
```
1

```
=ROUND(POWER(1.05,3),2)
```
1.16

Example: SQRT(50) = 7.0710678. ROUND(...,2) then cuts it to 7.07.

Nested Formula Rules

Read it from the inside out

Order

Excel solves the innermost function first, then works outward.

Brackets

Every function needs its own (and). Count them to avoid errors.

Arguments

The inner formula becomes one argument of the outer formula.

Debug

Evaluate Formula walks through each step. F9 shows a selected part (Esc to undo).

DAY 8 | EXCEL

Day 8 Shortcuts

Build and debug formulas

Evaluate Formula	Alt + M + V
Show value of selected part	F9
Cancel and keep the formula	Esc
Edit the formula	F2
Expand formula bar	Ctrl + Shift + U
Show all formulas	Ctrl + `
Insert Function box	Shift + F3
Lock a reference	F4

In the formula bar: select a part of the formula, press F9 to see its result, then press Esc so the formula is not replaced.

Interview Questions

Short and sharp answers

FLOOR vs ROUNDDOWN?

FLOOR goes down to a multiple (3, 5, 10). ROUNDDOWN cuts to a number of decimal places.

MROUND vs ROUND?

ROUND uses decimal places. MROUND uses the nearest multiple.

Check even or odd?

=MOD(A1,2): 0 means even, 1 means odd.

POWER vs ^?

Same result. 2^3 and POWER(2,3) both give 8.

Day 8 Cheat Sheet

All functions in one place

Multiples

FLOOR, CEILING, MROUND

Sign

ABS, SIGN

Division

MOD, QUOTIENT

Powers and Roots

POWER, ^, SQRT

Rounding Family

ROUND, ROUNDUP, ROUNDDOWN, INT, TRUNC

Day 8
Done!

FLOOR, CEILING, MOD, ABS,
POWER and SQRT.

Can you check even or odd with MOD?
Tell me in the comments.

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